

Differential Geometry Notes

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Definition 1. The first fundamental form of M at p is defined by restricting the standard inner product on $\mathbb{R}^k$ to the tangent space. We have $g_p : T_p M \times T_p M \rightarrow \mathbb{R}$

Definition 2. A diffeomorphism $F : M \rightarrow N$ is called an isometry if at each point the derivative DF_p is an isometry from $T_p M$ to $T_p N$

Definition 3. A smooth map $F : M \rightarrow N$ is called a local isometry at p if there is a neighborhood U such that $F(U)$ is open and $F|_U : U \rightarrow F(U)$ is an isometry

Claim 1. Two manifolds M and N are locally isometric around p and q if and only if there are local parametrizations $\phi : \Omega \rightarrow U$ and $\varphi : \Omega \rightarrow V$ such that $g_{\phi(a)}(v_i, v_j) = h_{\varphi(a)}(w_i, w_j)$ for all $a \in \Omega$

Definition 4. Suppose $X, Y \in \mathcal{X}(M)$ are smooth vector fields. Then we have the smooth vector field $\nabla_X Y$ given by $p \rightarrow \pi_p(DY_p(X_p))$

Claim 2. Suppose $F : M \rightarrow N$ is an isometry then $\nabla_{F_* X}(F_* Y) = F_*(\nabla_X Y)$

Definition 5. Suppose $\gamma : I \rightarrow M$ is a smooth curve. A smooth vector field along γ is a map $V : I \rightarrow \mathbb{R}^k$ such that $V(t) \in T_{\gamma(t)} M$

The vector space of smooth vector fields along γ is $\mathcal{X}(\gamma)$

Definition 6. We have $\text{cov}_\gamma \in L(\mathcal{X}(\gamma))$ given by $V \rightarrow (t \rightarrow \pi_{\gamma(t)} \dot{V}(t))$